

Decision and Social Choice

PE3101P — Lecture 2: Probability

Tomi Francis

National University of Singapore

20th August, 2026

Today

- 1 Logistics
- 2 Recap
- 3 Probabilities
- 4 Probabilism
- 5 Conditional probability
- 6 Dutch books

The week 1 exercise sheet is up

- ▶ On my website: <https://tomifrancis.com/pe3101p.html>

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- ▶ Optional, and not for credit.

A new document

I'm giving you the greatest gift a student can receive:

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- ▶ Optional — but may be useful for the for-credit assignment and other future material

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- ▶ In pdf and html. **I recommend the html.**

First for-credit assignment

- ▶ Goes out on my website **over the weekend**.

First for-credit assignment

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- ▶ Due **Friday of Week 4 — 4th September**, by midnight.

My office

- ▶ I'm moving office today. From now on I'm in **AS3-05-01**.

My office

- ▶ I'm moving office today. From now on I'm in **AS3-05-01**.
- ▶ **Office hours today are 16:30–18:00**, later than usual — I'm moving in right after this lecture.

The register

- ▶ Printed this time. **Find your name and sign.**

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- ▶ It's in alphabetical order.

The register

- ▶ Printed this time. **Find your name and sign.**
- ▶ It's in alphabetical order.
- ▶ Can't find yourself? Then I don't have records of you yet — add your name and student number, then sign.

Two senses of “ought”

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What will *in fact* be best.

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What your **evidence indicates** will be best, on the balance of probabilities.

- ▶ Run across the road without looking and survive: correct in the fact-relative sense, a mistake in the evidence-relative one.
- ▶ **Decision theory is about the evidence-relative sense.**

Acts, states, outcomes

	Miners in A	Miners in B
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- ▶ **States of nature:** the columns — ways the world could be, which you're uncertain about.
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Conditions on the states — what I gave you

- (i) Together with the act chosen, they **resolve the uncertainty** and guarantee one particular outcome.

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- (i) Together with the act chosen, they **resolve the uncertainty** and guarantee one particular outcome.
- (ii) The act you choose **does not affect** whether the state occurs, or how probable it is.

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These two are about the **set** of states, not each state on its own.

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Like rolling a die: one number ends up face up.

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- ▶ But **objective chances** are different: they're out in the world.
- ▶ Frequencies, or what a theory like quantum mechanics puts out.
- ▶ **Credences are in the agent's head** — how likely she thinks something is, given her evidence.

Utility and expected utility

A fair coin. **Take the bet:** \$100 on heads, nothing on tails.

Decline: \$50 either way.

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- ▶ Where do the numbers come from? Later in the course. For now: **more is better**.
- ▶ The **expected utility** of an act: the utilities of its outcomes, weighted by the probabilities of the states.

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Strong statewise dominance

Better in **all** states.

Bayesianism: two constraints

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Today: both of them.

The sample space

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A sample space is just a **set**: a collection of things.

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Events

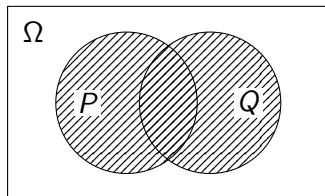
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- ▶ The event that *anything at all* happens is Ω itself. Last week I called this the **universal event**.
- ▶ The event with **no** outcomes in it is $\emptyset$: the **empty event**. It can never happen.

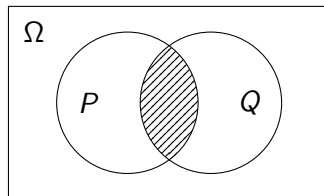
Combining events

$P \cup Q$ — something in **either** happens



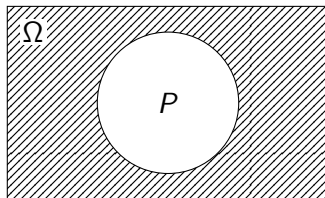
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Combining events

P^c — everything in Ω **outside** P



We'll often write $\neg P$ for this.

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I won't penalise you for writing $\wedge$ instead of $\cap$.

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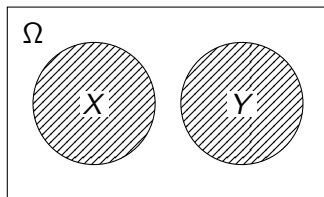
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We'll assume any set of outcomes is an event.

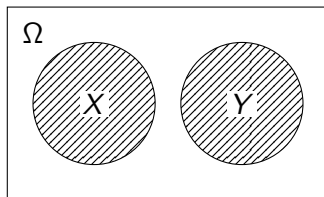
Mutually exclusive

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Also called **disjoint**.

Partitions

Jointly exhaustive: $X_1 \cup \dots \cup X_n = \Omega$

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X_1	X_2	X_4

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Exactly one of them obtains — like the states of a decision table.

What's wrong with these?

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Homer thinks there's a 30% chance he'll drink one beer, a 50% chance he'll drink more than one, and a 70% chance he'll drink at least one.

The axioms — informally, as last week

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A die lands 1 *or* 2 with probability $1/6 + 1/6 = 1/3$.

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The axioms

- (i) $\text{Cr}(\sigma) \geq 0$ for every $\sigma \in \Sigma$
- (ii) $\text{Cr}(\Omega) = 1$
- (iii) if $X \cap Y = \emptyset$, then $\text{Cr}(X \cup Y) = \text{Cr}(X) + \text{Cr}(Y)$

So what went wrong?

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- ▶ **Biqing.** Failing badly is a *way of* failing. So it can't be more probable than failing.
- ▶ **Homer.** One beer and more than one are disjoint, and together they are “at least one”. So that should be $0.3 + 0.5 = 0.8$ — not 0.7.

Conditional probability

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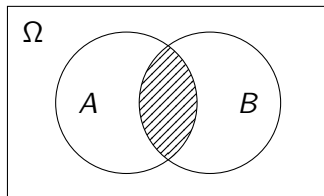
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Your credence in A , **on the supposition that B** .

Note the proviso: conditioning on something you've ruled out is **undefined**, not zero.

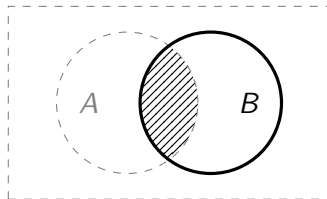
Why that formula?

Before: Ω is everything.



Why that formula?

After: B is everything.



The shaded region is $A \cap B$ both times. What changes is **what it's measured against**.

Probabilistic independence

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Learning B tells you **nothing** about A .

Conditionalisation

The Principle of Conditionalisation

Suppose your credences are Cr , and you learn E **and nothing more**. Then your new credence in any P should be

$$Cr_{\text{new}}(P) = Cr(P \mid E).$$

Conditionalisation

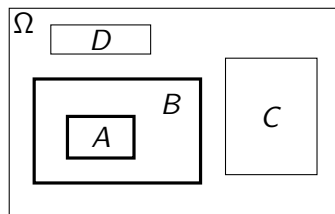
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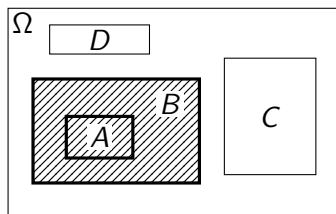
Your old **conditional** credence becomes your new **unconditional** one.

Conditionalisation, in pictures



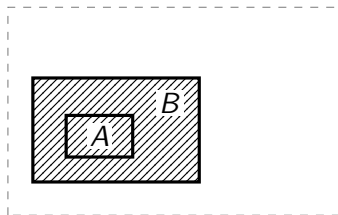
Areas are probabilities. We're going to condition on B .

Conditionalisation, in pictures



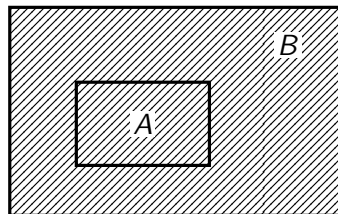
B is the **conditioning event**.

Conditionalisation, in pictures



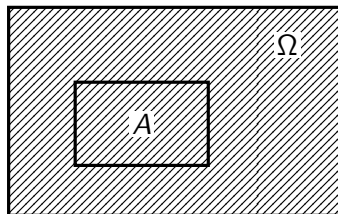
Everything outside B is **discarded**. Nothing has moved yet.

Conditionalisation, in pictures



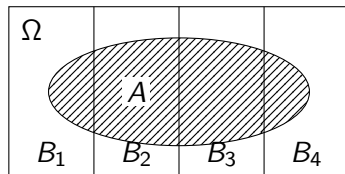
B **blown up** to fill the space Ω used to. Every ratio inside is unchanged.

Conditionalisation, in pictures

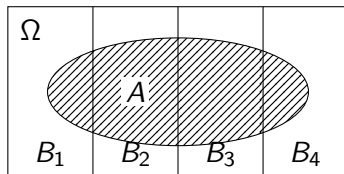


B is the new sample space.

Total probability

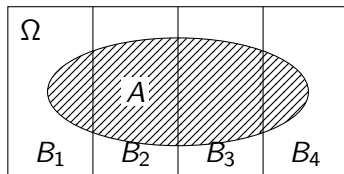


Total probability



The B_i **partition** Ω , so they cut A into disjoint pieces $A \cap B_i$.

Total probability



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Their probabilities **add up to** $\text{Cr}(A)$.

Bayes' theorem: what we have

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Bayes factors

Two hypotheses H_1 , H_2 , and some evidence E .

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A Bayes factor of 1 moves you not at all.

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- ▶ An urn holds 3 red and 7 black. Two drawn without replacement. Credence the second is red, given the first was red? And before you look at the first?
- ▶ $\text{Cr}(A) = 0.5$, $\text{Cr}(B) = 0.4$, $\text{Cr}(A \cap B) = 0.2$. Find $\text{Cr}(A \mid B)$ and $\text{Cr}(B \mid A)$. **Are A and B independent?**

Base rates

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"So there's only a one in a million chance he's innocent."

Linking belief to action

Write a **\$100 bet on A** for a deal paying \$100 if A is true, nothing otherwise.

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We'll assume this today. It's a big assumption. Complaints held over.

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She loses \$40 **whatever happens**.

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And if $\text{Cr}(\Omega) = 1.2$? She buys the same bet for \$110 and collects \$100. **Same \$10 loss.**

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So we'll need **more than one bet**.

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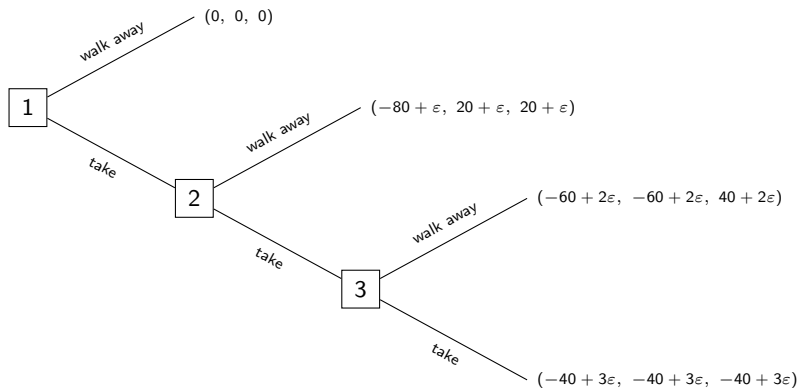
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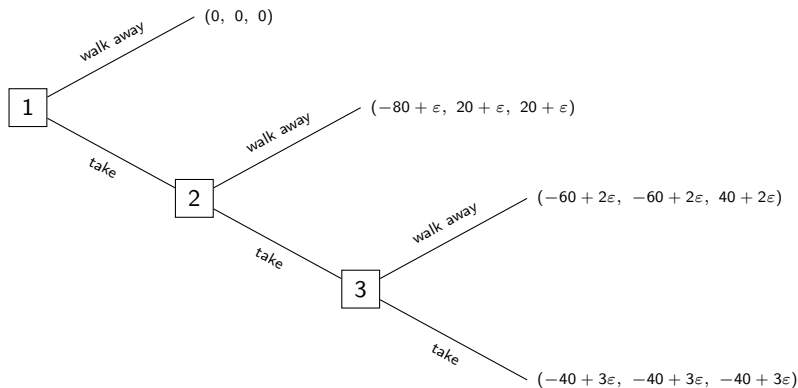
Bet 3 She **buys** a \$100 bet on $P \cup Q$ for $\$80 - \varepsilon$. Her price is \$80, so buying below it is a gain.

Each one, on its own, is a bet she **wants**.

But take all three...



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She is down about **\$40**, however things turn out.

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So she walks away at node 1.

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Week 5, if there's time.

From the theorem to an argument

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- ▶ But there might equally be a **Frenchman** going around **rewarding** people with non-probabilistic credences.
- ▶ We wouldn't conclude you ought to have them. At most, that having them would be **practically useful**.
- ▶ And that's the problem: losing money is a **practical** misfortune. What was at issue is whether your beliefs are **epistemically** in order.

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- 1 You are rationally required to bet in line with the betting interpretation, given the credences you have.

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- 1 You are rationally required to bet in line with the betting interpretation, given the credences you have.
 - 2 If you have non-probabilistic credences and bet in line with the betting interpretation, you are vulnerable to a Dutch book.
 - 3 If you are vulnerable to a Dutch book, your credences are irrational.
-

So: if your credences are non-probabilistic, they are **irrational**.

The Frenchman

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Making choices which **predictably** leave you worse off, for no reason, is irrational.

And it isn't about money

You're not rationally required to value money.

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The argument works for **any** quantity you do care about — so long as you're required to bet in line with the betting interpretation for it.

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- ▶ There's an answer: it's much more plausible when the stakes are **small**.
- ▶ And however small you make the stakes, you can **still** be made to lose for sure.

For next week

Newcomb's problem

BDRC ch. 9, and Adam Elga, “Newcomb University: A Play in One Act”, *Analysis* 80.2 (2020), pp. 212–221.

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See you next week.